

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2,BL3)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Case Study

Duration: 2hrs

1. A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot’s learning and behavior. (5 Marks)

RL Components (MDP)

Component	Description
States	Robot location, item location, battery level, obstacle positions, delivery status
Actions	Move Up, Down, Left, Right, Pick Item, Drop Item, Wait
Environment	Warehouse with shelves, packages, obstacles, and charging station
Reward	+20 for successful delivery, -10 for collision, -2 for delay, +5 for correct pick-up
Policy	Choose the shortest and safest path to complete deliveries

Reward Function

$$\text{Reward} = +20(\text{Delivery}) + 5(\text{Correct Pick}) - 10(\text{Collision}) - 2(\text{Delay})$$

Analysis

- Positive rewards encourage faster and accurate deliveries.
- Negative rewards discourage collisions and unnecessary delays.
- The agent learns efficient warehouse navigation.

Evaluation

A well-designed reward function improves speed, safety, and accuracy, leading to efficient warehouse operations with fewer collisions.

2. A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation. (5 Marks)

RL Components

Component	Description
States	User age, viewing history, watch time, ratings, preferred genres
Actions	Recommend a movie from the available catalog
Environment	Streaming platform and user interactions
Reward	+5 if the movie is watched completely, +3 for a like, -3 if skipped
Policy	Recommend movies with the highest expected user satisfaction

Analysis

- Positive rewards increase when users watch recommended movies.
- Negative rewards occur when recommendations are skipped.
- The system continuously adapts to changing user preferences.

Exploration vs Exploitation

- Exploration: Recommend new genres.
- Exploitation: Recommend movies similar to previous favorites.

Evaluation

Balancing exploration and exploitation improves user engagement while avoiding repetitive recommendations.

4.A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making. (5 Marks)

RL Components

Component	Description
States	Network traffic, login attempts, packet information, system status
Actions	Allow traffic, Block traffic, Raise alert, Request authentication
Environment	Enterprise computer network
Reward	+10 for detecting attacks, -15 for allowing attacks, -5 for false alarms
Policy	Detect attacks while minimizing false positives

Analysis

- Sparse rewards occur because attacks are relatively rare.
- The agent must make fast decisions with limited information.
- Real-time learning is required to handle evolving threats.

Evaluation

RL enables adaptive intrusion detection, but sparse rewards and changing attack patterns make learning challenging.

6.A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. (5 Marks)

RL Components

Component	Description
States	Room occupancy, temperature, humidity, weather, time of day
Actions	Turn lights ON/OFF, Adjust AC/heater, Control fan speed
Environment	Smart home with sensors and appliances
Reward	+10 for comfort, +8 for energy saving, -5 for unnecessary power usage
Policy	Maintain comfort while minimizing electricity consumption

Analysis

The system has two objectives:

- User comfort
- Energy efficiency

Reward shaping combines both objectives into a single reward function, helping the agent learn balanced behavior.

Evaluation

Reward shaping improves learning speed and allows the system to provide comfort without excessive energy consumption.

10.A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this context. Design a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning. (5 Marks)

RL Components (MDP)

Component	Description
States	Current electricity demand, renewable energy availability, battery storage, time
Actions	Increase supply, Reduce supply, Store energy, Use battery backup
Environment	Smart electricity distribution network
Reward	+15 for balanced supply, -10 for overload, -5 for energy wastage
Policy	Distribute electricity efficiently while maintaining grid stability

Reward Function

$$\text{Reward} = +15(\text{Balanced Supply}) - 10(\text{Overload}) - 5(\text{Energy Waste})$$

Analysis

The MDP helps the system make sequential decisions under changing electricity demand while maintaining reliability.

Evaluation

Challenges include:

- Large-scale power networks (scalability)
- Real-time decision-making
- Fluctuating renewable energy generation

RL enables efficient electricity distribution while reducing peak loads and improving reliability.

Code of Conduct:

I _S.Sirivally/ 192472371_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sirivally.S

RUBRICS FOR CASESTUDY

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand